Christopher Turner

Lead Engineer | Applied AI · Clickstream Data · Product Discovery

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Professional Summary

Lead engineer with 17+ years' experience, working from raw data through to shipped product. At Alterian I work directly with high-volume raw clickstream data: building the pipelines, finding patterns in how customers actually behave, and using that to decide what is worth building. I then lead the team that builds it, including production LLM features on AWS Bedrock shipped to real customers.

As much product discovery as delivery. Much of my time goes on the questions before the code: exploring the data, testing ideas with product and commercial teams, prototyping, and prioritising. I can then build end to end, across Go and TypeScript backends, Angular and server-rendered HTMX/Alpine front-ends, and AWS-native infrastructure.

Strong record of mentoring developers, modernising legacy stacks, and turning manual delivery processes into automated CI/CD pipelines. Equally happy in a Lead/Principal role setting direction or as a Senior IC building the hard parts.

Experience

Senior Web Developer ➡ Lead Web Developer

Alterian (2015 to Present)

- **Product Discovery & Clickstream Data:** Work from raw, high-volume clickstream data upwards: building the ingestion and preprocessing pipelines (**Parquet** queried with **DuckDB**, and **StarRocks** as the analytical database), digging into it to find patterns in real customer journeys, and turning those findings into product hypotheses, prototypes and feature priorities with product and commercial teams.
- **AI, Search & LLM Integration:** Designed and shipped production LLM features on **AWS Bedrock**, including conversational insights over large-scale customer-journey datasets, with a **tool-calling architecture** that lets the model query internal APIs and data stores (SQL, Parquet/DuckDB) rather than stuffing raw data into prompts. Now applying **embeddings** and **vector similarity search** to classify and search raw journey data, and building a comparison tool that combines several retrieval methods with an **LLM-as-judge** in the loop to score and rank results.
- **Technical Architecture & Modernisation:** Led two major platform transformations. First, unified the web stack on **Angular**, **Material Design**, **Nx monorepos**, **Tailwind CSS**, and **Storybook** with atomic design principles. More recently, led a strategic move to **Go** with lightweight HTMX/Alpine front-ends for new products, delivering faster CI/CD, lower hosting cost, and a smaller surface area for the team to maintain.
- **Leadership & Team Development:** Provide technical leadership and mentoring across the development team, guiding Junior Developers through to Senior roles and growing UI specialists into full-stack engineers. Run code reviews, design sessions, and architecture discussions, and bridge engineering, product and commercial teams by translating business needs into technical direction.
- **DevOps & Cloud Infrastructure:** Led the migration from manual 6-weekly deployments to automated CI/CD using **Docker**, **ECS Fargate**, and **Bitbucket Pipelines**, giving the team complete deployment control and continuous delivery. Built cloud-native architecture across the AWS ecosystem (**ECS, Lambda, S3, EventBridge, DynamoDB, Cognito, Bedrock**) with **Terraform**, and established secure development practices (SSDLC) and a comprehensive testing strategy.

System Tester ➡ DevOps ➡ Senior Web Developer

Computershare (2008 to 2015)

Progressed from system testing through DevOps into senior web development across multiple products. Championed mobile-first and progressive-enhancement strategies for legacy applications, introduced test automation that reduced QA cycle times, and worked directly with business clients on demos, requirements gathering, and wireframing. Gained 3rd-line support experience and contributed to feature development, technical risk assessment, and resource planning.

Technical Stack

Currently building with: Go, TypeScript, HTMX/Alpine, Angular, Tailwind, AWS (Bedrock, Lambda, ECS), Terraform, Docker.

Data: Raw clickstream processing, Parquet, DuckDB, StarRocks, SQL, data visualisation.

AI & search: Production Bedrock features with tool use and structured outputs; embeddings, vector similarity search, classification, and LLM-as-judge evaluation.

Comfortable in: React, Node.js, PHP, MySQL, Nx monorepos, Storybook, Figma.

Earlier career: C#, .NET, ASP.NET, SQL Server.

Projects

Job-Search Agent

Go · local LLMs (LM Studio) · MCP

- **Built an LLM-powered agent** that pulls postings from Hacker News "Who is hiring?", scores each against a structured criteria profile, and produces a ranked, explained shortlist.
- **Schema-constrained JSON output** at temperature 0 for deterministic classification, behind pluggable source and LLM-backend interfaces so a new source or model is a single new file.
- **Exposed as an MCP server**, so its tools can be used from Claude Desktop as well as the CLI.

Bristol Rowing

bristolrowing.co.uk — **Designer, Developer, and Maintainer**

- **Built and maintain a full rowing-club website** with Angular front-end and PHP/MySQL back-end.
- **Delivered a CMS** that lets the communications team manage public content independently.
- **Developed the membership portal**, covering resource management, bookings, treasury/finance, and event coordination.

Education

Bristol University (2005 to 2008)

Computer Science (BSc) — 2:1

Chislehurst and Sidcup Grammar School (1998 to 2005)

GCSEs and A Levels

Additional Leadership Experience

Trustee, City of Bristol Rowing Club (2013 to 2019)

Contributed to strategic direction and operational guidance for a charity with a large membership and substantial annual turnover.

Head Coach, City of Bristol Rowing Club Women's Squad (2016 to 2021)

Led a performance sports team through measurable growth in both participation and competitive results. Built a culture focused on individual development and process-driven improvement, balancing competitive goals with athlete wellbeing.